22 A source of sound waves is moving at a constant speed directly towards a stationary observer.

The sound waves have a speed of $340\,m\,s^{-1}$ and a frequency of 480 Hz. The observer hears sound waves of frequency 650 Hz.

What is the speed of the source?

A $89\,m\,s^{-1}$ **B** $120\,m\,s^{-1}$ **C** $250\,m\,s^{-1}$ **D** $340\,m\,s^{-1}$